Keep all tiles flat and in one piece, joined along whole sides without overlaps. Count every side with no tile beside it, including sides around holes. Turns and flips count as the same shape.

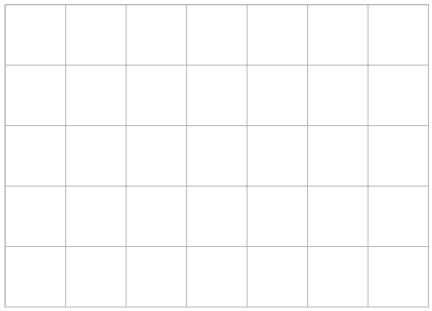
Problem 1: Find the shortest boundary you can with 7, 10, and 13 tiles. For each number, find two different shapes reaching your best length, if you can.



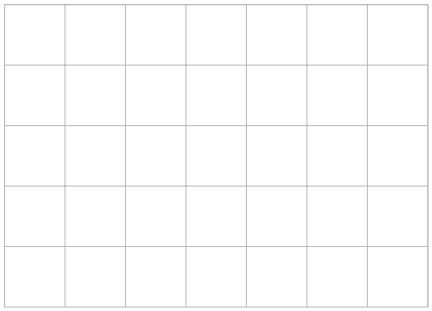
7 tiles; boundary: _____



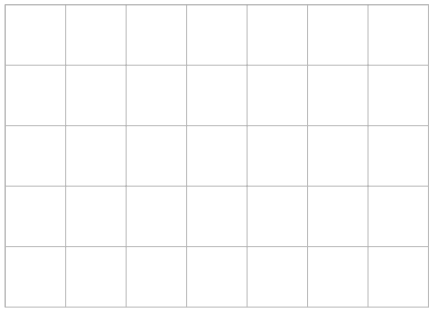
7 tiles; boundary: _____



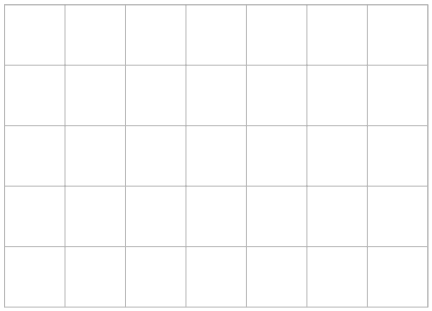
10 tiles; boundary: _____



10 tiles; boundary: _____

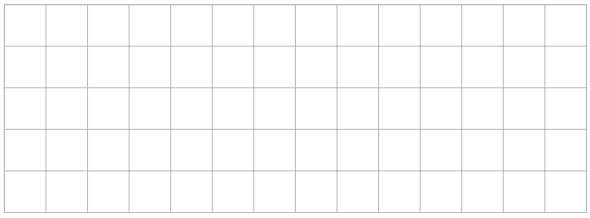


13 tiles; boundary: _____

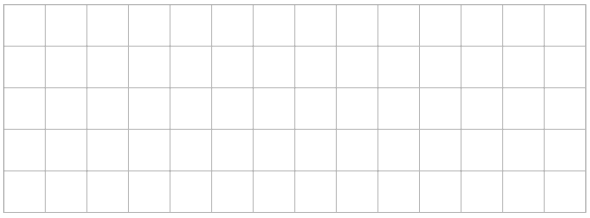


13 tiles; boundary: _____

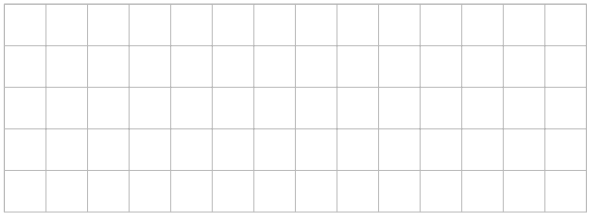
Problem 2: What is the longest possible boundary of a twelve-tile shape? Explain why no shape can have a longer boundary. Draw three shapes reaching that length, none of them a straight row. Can such a shape contain a 2-by-2 block? Can it have a hole? Two tiles that touch at a corner leave no gap between them.



Boundary: _____

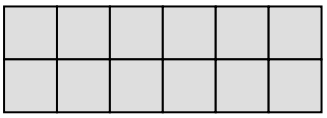


Boundary: _____



Boundary: _____

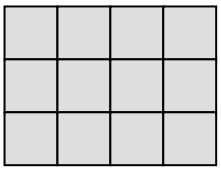
Problem 3: Count the occupied rows, occupied columns, and boundary sides of each shape.



Rows: ____

Columns: ____

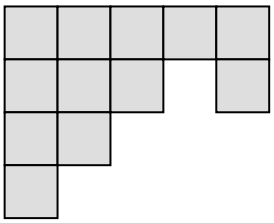
Boundary: ____



Rows: ____

Columns: ____

Boundary: ____

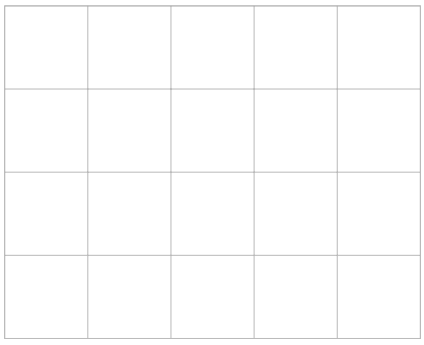


Rows: ____

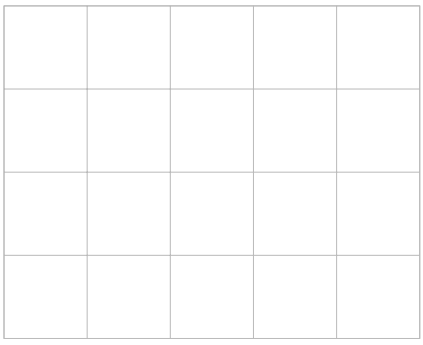
Columns: ____

Boundary: ____

Make two twelve-tile shapes that each occupy 4 rows and 5 columns but have different boundary lengths. What is the least possible boundary length for a shape occupying 4 rows and 5 columns? Explain.

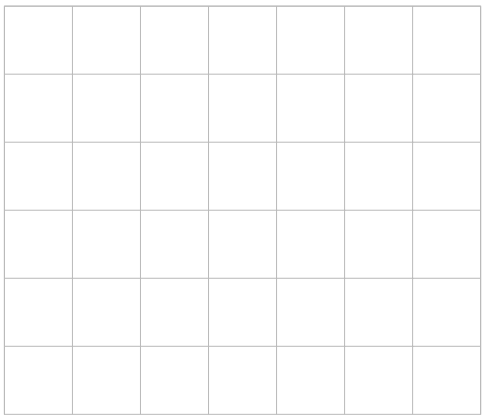


Boundary: _____

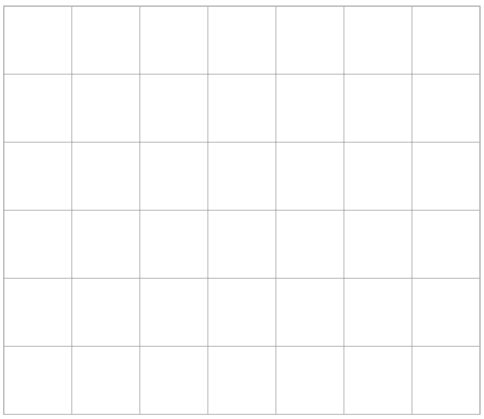


Boundary: _____

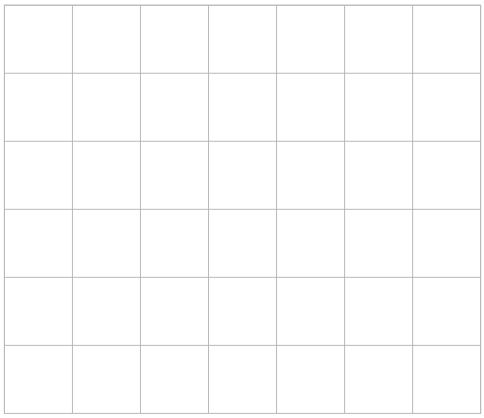
Problem 4: For each boundary length shown, find the greatest number of tiles a shape can have. Explain why no shape with that boundary length can contain more tiles.



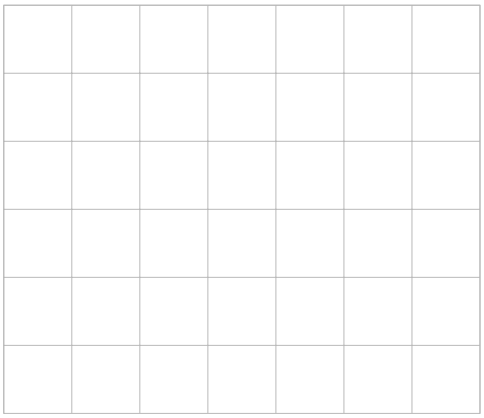
Boundary: 12; tiles: _____



Boundary: 14; tiles: _____

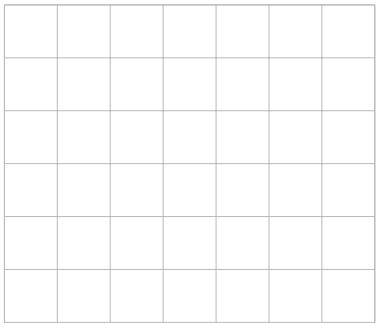


Boundary: 16; tiles: _____

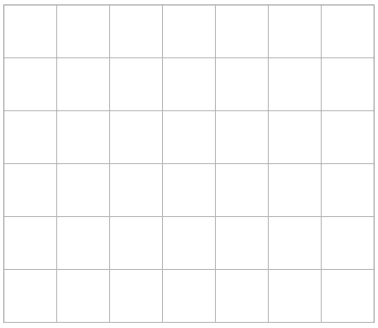


Boundary: 18; tiles: _____

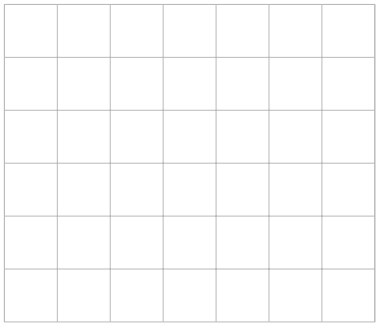
Problem 5: Find the shortest possible boundary for shapes with 12, 13, 17, 20, and 21 tiles. Explain why each of your shapes has the shortest possible boundary.



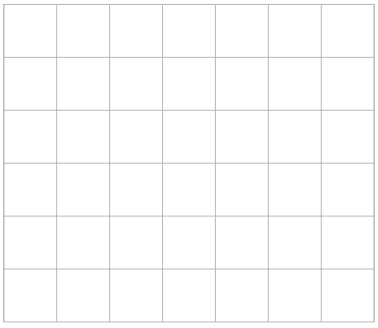
12 tiles; boundary: _____



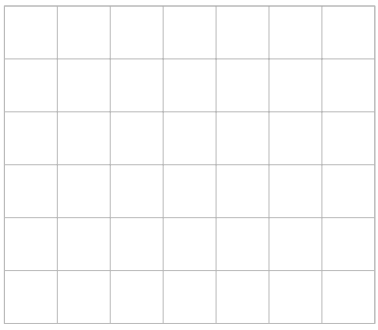
13 tiles; boundary: _____



17 tiles; boundary: _____

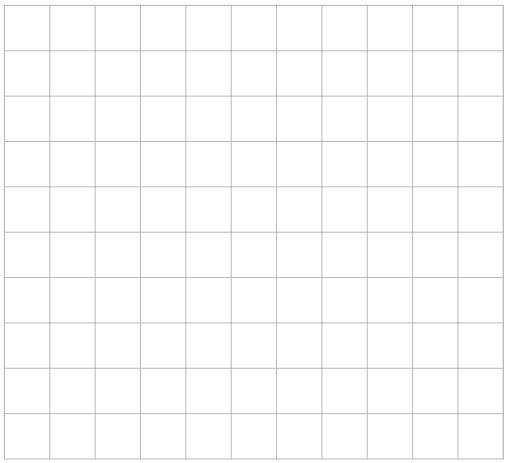


20 tiles; boundary: _____

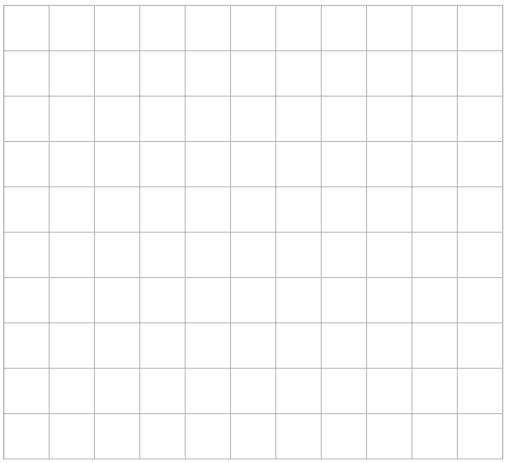


21 tiles; boundary: _____

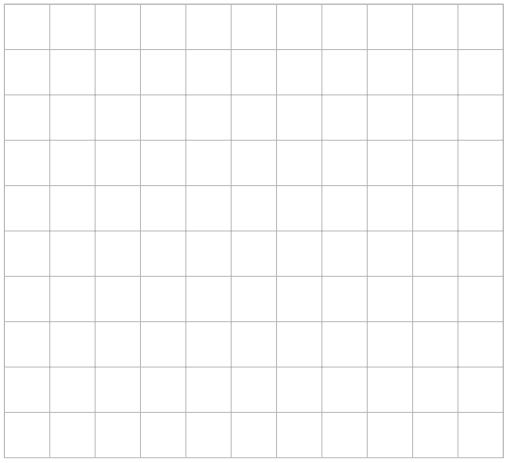
Problem 6: Give a rule for the greatest number of tiles in a shape with any even boundary length of at least 4. Explain why your rule works. Use it to find the shortest possible boundaries for shapes with 37, 50, and 73 tiles, and draw shapes that reach them.



37 tiles; boundary: _____



50 tiles; boundary: _____



73 tiles; boundary: _____